VIT CHENNAI · BMAT202L

Probability
& Statistics

Complete Study Guide · 7 Modules · Formula Sheets · Solved Problems

- Module 1: Descriptive Statistics & Correlation
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MODULE 1

Descriptive Statistics & Correlation

Descriptive statistics summarise data with measures of central tendency (mean, median, mode) and dispersion (range, variance, SD). Correlation and regression measure the strength and direction of linear relationships between two variables.

1.1 Measures of Central Tendency & Dispersion

DESCRIPTIVE STATISTICS FORMULAS

ARITHMETIC MEAN

$$\bar{x} = \frac{\sum x_i}{n} = \frac{\sum f_i x_i}{\sum f_i}$$

VARIANCE (POPULATION)

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x_i^2}{n} - \bar{x}^2$$

VARIANCE (SAMPLE)

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

COEFFICIENT OF VARIATION

$$\mathrm{CV} = \frac{\sigma}{\bar{x}} \times 100\%$$

SKEWNESS (KARL PEARSON)

$$S_k = \frac{\bar{x} - \mathrm{Mode}}{\sigma}$$

MEDIAN (GROUPED)

$$M = L + \frac{N/2 - F}{f} \times h$$

1 Variance and SD

Find mean, variance, and SD of: 2, 4, 4, 6, 8, 8, 8, 10.

SOLUTION

1. $\bar{x} = \frac{2+4+4+6+8+8+8+10}{8} = \frac{50}{8} = 6.25$
2. $\sum x_i^2 = 4 + 16 + 16 + 36 + 64 + 64 + 64 + 100 = 364$
3. $\sigma^2 = 364/8 - 6.25^2 = 45.5 - 39.0625 = 6.4375.\ \sigma \approx 2.54$

Mean = 6.25, Variance ≈ 6.44, SD ≈ 2.54

1.2 Correlation & Regression

CORRELATION & REGRESSION FORMULAS

PEARSON'S R

$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n\sigma_x\sigma_y} = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 -}}$

REGRESSION Y ON X

$y - \bar{y} = b_{yx}(x - \bar{x})$ where $b_{yx} = r\frac{\sigma_y}{\sigma_x}$

REGRESSION X ON Y

$x - \bar{x} = b_{xy}(y - \bar{y})$ where $b_{xy} = r\frac{\sigma_x}{\sigma_y}$

R FROM REGRESSION COEFF.

$r^2 = b_{yx} \cdot b_{xy}$ (both same sign as r)

SPEARMAN'S RANK CORR.

$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$

|R| INTERPRETATION

0: no corr; 0.5–0.7: moderate; 0.7–0.9: strong; >0.9: very strong

1 Pearson's correlation coefficient

For n=5, Σx=15, Σy=25, Σx²=55, Σy²=135, Σxy=83. Find r.

SOLUTION

1. Numerator: $n\Sigma xy - \Sigma x \Sigma y = 5(83) - 15(25) = 415 - 375 = 40$
2. $D_x = n\Sigma x^2 - (\Sigma x)^2 = 5(55) - 225 = 275 - 225 = 50$
3. $D_y = n\Sigma y^2 - (\Sigma y)^2 = 5(135) - 625 = 675 - 625 = 50$
4. $r = 40/\sqrt{50 \times 50} = 40/50 = 0.8$

r = 0.8 (strong positive correlation)

2 Spearman's rank correlation

Ranks: d values = 1, −2, 0, 1, 2 for n=5. Find r_s.

SOLUTION

1. $\sum d_i^2 = 1 + 4 + 0 + 1 + 4 = 10$
2. $r_s = 1 - \frac{6(10)}{5(24)} = 1 - \frac{60}{120} = 0.5$

$r_s = 0.5$ (moderate positive correlation)

Module 1 — Key Formulas

Variance

$\sigma^2 = \Sigma x_i^2/n - \bar{x}^2$

Pearson r

$b_{yx} = r\sigma_y/\sigma_x$

Spearman r_s

$1 - 6\Sigma d^2/[n(n^2 - 1)]$

MODULE 2

Probability & Random Variables

Probability theory provides the mathematical language for uncertainty. Random variables formalise outcomes numerically; expectation and variance capture the average behaviour and spread, forming the backbone of statistical inference.

2.1 Probability Axioms & Bayes' Theorem

PROBABILITY LAWS

BAYES' THEOREM

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

TOTAL PROBABILITY

$$P(B) = \sum_i P(B|A_i)P(A_i)$$

INDEPENDENCE

$$P(A \cap B) = P(A) \cdot P(B)$$

1 Bayes' Theorem

A test is 95% accurate for disease (D). Disease prevalence = 1%. P(D|positive)?

SOLUTION

1. $P(D)=0.01$, $P(+|D)=0.95$, $P(+|\bar{D})=0.05$ (false positive)
2. $P(+)=0.95(0.01)+0.05(0.99)=0.0095+0.0495=0.059$
3. $P(D|+)=0.0095/0.059\approx 0.161$

$P(D|+) \approx 16.1\%$ (surprisingly low due to low prevalence)

2.2 Random Variables, PDF, CDF, E(X), Var(X)

RANDOM VARIABLE FORMULAS

CDF (CONTINUOUS)

$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$

EXPECTATION E(X)

$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$ (continuous)

VARIANCE VAR(X)

$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$

E(AX+B)

$aE(X) + b$

VAR(AX+B)

$a^2\mathrm{Var}(X)$

MOMENT GENERATING FUNCTION

$M_X(t) = E(e^{tX})$

1 Find E(X) and Var(X)

Let $f(x) = 2x$ for $0 \le x \le 1$. Find E(X) and Var(X).

SOLUTION

1. $E(X) = \int_0^1 x \cdot 2x\,dx = 2\int_0^1 x^2\,dx = 2/3$
2. $E(X^2) = \int_0^1 x^2 \cdot 2x\,dx = 2\int_0^1 x^3\,dx = 1/2$
3. $\mathrm{Var}(X) = 1/2 - (2/3)^2 = 1/2 - 4/9 = 1/18$

E(X) = 2/3 ≈ 0.667; Var(X) = 1/18 ≈ 0.056

Module 2 — Key Formulas

Bayes'

$P(A|B) = P(B|A)P(A)/P(B)$

E(X)

$\int x f(x) dx$

Var(X)

$E(X^2) - [E(X)]^2$

MODULE 3

Probability Distributions

Standard probability distributions model real-world phenomena: Binomial for coin flips and defects, Poisson for rare events and arrival rates, Normal for measurement errors and heights, Exponential for lifetimes and waiting times.

3.1 Discrete Distributions

DISCRETE DISTRIBUTIONS

BINOMIAL B(N,P)

$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$

E(X)=np; Var=npq

POISSON P(Λ)

$P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}$

E(X)=λ; Var(X)=λ

BINOMIAL→POISSON

When n large, p small, np=λ constant

1 Binomial Distribution

If a die is thrown 5 times, find P(exactly 2 sixes).

SOLUTION

1. n=5, k=2, p=1/6, q=5/6
2. $P(X=2)=\binom{5}{2}(1/6)^2(5/6)^3=10\cdot\frac{1}{36}\cdot\frac{125}{216}=\frac{1250}{7776}\approx 0.1608$

P(X=2) ≈ 0.1608 ≈ 16.08%

2 Poisson Distribution

Calls arrive at 3 per minute. Find P(exactly 5 calls in 1 minute).

SOLUTION

1 λ=3. $P(X=5)=\frac{e^{-3}\cdot 3^5}{5!}=\frac{0.0498\cdot 243}{120}=\frac{12.1}{120}\approx 0.1008$

P(X=5) ≈ 0.1008 ≈ 10.08%

3.2 Continuous Distributions

CONTINUOUS DISTRIBUTIONS

NORMAL N(Μ,Σ²)

$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$; Standard: $Z=\frac{X-\mu}{\sigma}$

EXPONENTIAL (Λ)

$f(x)=\lambda e^{-\lambda x},\ x\geq 0$; E(X)=1/λ; Var=1/λ²

GAMMA Γ(Α,Β)

$f(x)=\frac{x^{\alpha-1}e^{-x/\beta}}{\beta^\alpha\Gamma(\alpha)}$; E(X)=αβ; Var=αβ²

WEIBULL W(Α,Β)

$f(x)=\frac{\alpha}{\beta^\alpha}x^{\alpha-1}e^{-(x/\beta)^\alpha}$

NORMAL: 68-95-99.7 RULE

μ±σ: 68%; μ±2σ: 95%; μ±3σ: 99.7%

MEMORYLESS PROPERTY

Exponential: P(X>s+t|X>s)=P(X>t)

1 Normal distribution

X~N(50, 25). Find P(45 < X < 60).

SOLUTION

1 $\sigma=5.\ Z_1=(45-50)/5=-1;\ Z_2=(60-50)/5=2$

2 P(−1<Z<2)=Φ(2)−Φ(−1)=0.9772−0.1587=0.8185

P(45 < X < 60) ≈ 0.8185 ≈ 81.85%

Module 3 — Key Formulas

Binomial

Poisson

Normal Z-score

$E=np$; $Var=npq$	$E=Var=\lambda$	$Z = (X - \mu)/\sigma$

MODULE 4

Sampling Theory & Confidence Intervals

The Central Limit Theorem is the cornerstone of inferential statistics: sample means are approximately normally distributed regardless of the population distribution, enabling us to construct confidence intervals and conduct hypothesis tests with known error rates.

4.1 Central Limit Theorem & Standard Error

CENTRAL LIMIT THEOREM (CLT)

For a random sample of size n from any population with mean μ and variance σ^2:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ approximately (for large n)}$$

Standard error of the mean: $\mathrm{SE} = \frac{\sigma}{\sqrt{n}}$

SAMPLING DISTRIBUTIONS

Z-STATISTIC (Σ KNOWN)

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

T-STATISTIC (Σ UNKNOWN)

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}},\ df = n - 1$$

SAMPLE PROPORTION SE

$$\mathrm{SE}(\hat{p}) = \sqrt{p(1-p)/n}$$

1 CLT problem

Population: μ=100, σ=20. Sample n=64. Find P($\bar{X}$ > 105).

SOLUTION

1. SE = σ/√n = 20/8 = 2.5. Z=(105−100)/2.5=2
2. P($\bar{X}$>105)=P(Z>2)=1−Φ(2)=1−0.9772=0.0228

P(X̄ > 105) ≈ 0.0228 ≈ 2.28%

4.2 Confidence Intervals

CONFIDENCE INTERVAL FORMULAS

CI FOR M (Σ KNOWN, 95%)

$\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$

CI FOR M (Σ UNKNOWN)

$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$

CI FOR PROPORTION P

$\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$

COMMON Z VALUES

90%: z=1.645; 95%: z=1.96; 99%: z=2.576

WIDTH OF CI

Width = 2 × margin of error; wider = less precise

SAMPLE SIZE FOR MARGIN E

$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$

1 Confidence interval for μ

Sample: n=36, x̄=45, s=6. Construct 95% CI for μ.

SOLUTION

1. Since n=36 (large), use z=1.96. SE=s/√n=6/6=1
2. CI: $45 \pm 1.96 \times 1 = 45 \pm 1.96 = (43.04,\ 46.96)$

95% CI: (43.04, 46.96)

2 t-interval (small sample)

n=10, x̄=72, s=8. Find 95% CI ($t_{0.025,9}$ = 2.262).

SOLUTION

1. SE=8/√10=2.53. Margin=2.262×2.53=5.72

2 CI: (72−5.72, 72+5.72)=(66.28, 77.72)

95% CI: (66.28, 77.72)

Module 4 — Key Formulas

CLT	SE	95% CI (σ known)
$\bar{X} \sim N(\mu, \sigma^2/n)$	$\sigma/\sqrt{n}$	$\bar{x} \pm 1.96 \cdot \mathrm{SE}$

MODULE 5

Hypothesis Testing

Hypothesis testing provides a formal procedure for deciding whether evidence from a sample contradicts a null hypothesis. The Z-test, t-test, χ^2 test, and F-test each suit different scenarios based on sample size, known parameters, and data type.

5.1 Framework & Z/t Tests

HYPOTHESIS TESTING FRAMEWORK

H_0 AND H_1

H_0: null hypothesis (status quo); H_1: alternative (claim to test)

TYPE I ERROR (Α)

Reject H_0 when true (significance level)

TYPE II ERROR (Β)

Fail to reject H_0 when false

Z-TEST (Σ KNOWN / LARGE N)

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

T-TEST (Σ UNKNOWN, SMALL N)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},\ df = n - 1$$

TWO-SAMPLE T-TEST

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}},\ df = n_1 + n_2 - 2$$

1 One-sample Z-test

Claim: $\mu=50$. Sample: $n=100$, $\bar{x}=52$, $\sigma=10$. Test at $\alpha=0.05$ (two-tailed).

SOLUTION

1. H_0: $\mu=50$; H_1: $\mu\neq50$. $Z=(52-50)/(10/\sqrt{100})=2/1=2$
2. Critical value for two-tailed $\alpha=0.05$: z_crit=1.96. $|Z|=2>1.96$
3. Reject H_0. Evidence that $\mu\neq50$.

Reject H_0 ($Z=2 > 1.96$ at $\alpha=0.05$)

5.2 Chi-square Test & F-test

χ² AND F-TEST

χ² GOODNESS-OF-FIT

$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}, \ df = k - 1$

χ² TEST OF INDEPENDENCE

df=(r−1)(c−1); $E_{ij} = \frac{\mathrm{Row}_i \times \mathrm{Col}_j}{N}$

F-TEST (VARIANCE RATIO)

$F = s_1^2/s_2^2$ where $s_1^2 > s_2^2$; $df_1=n_1-1$, $df_2=n_2-1$

1 Chi-square goodness-of-fit

Observed: O=[20,30,25,25]; Expected: E=[25,25,25,25] (uniform). Test at α=0.05.

SOLUTION

1. $\chi^2 = \frac{(20-25)^2}{25} + \frac{(30-25)^2}{25} + \frac{(25-25)^2}{25} + \frac{(25-25)^2}{25} = \frac{25+25}{25} = 2$
2. df=k−1=3. χ²_crit(0.05, 3)=7.815. χ²=2 < 7.815
3. Fail to reject H_0. Data fits uniform distribution.

Fail to reject H_0 ($\chi^2=2 < 7.815$). Data is uniformly distributed.

Module 5 — Key Formulas

Z-test	χ² test	Type I Error α
$Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	$\sum(O-E)^2/E$	Reject true H_0

MODULE 6

ANOVA & Reliability Theory

Analysis of Variance (ANOVA) tests equality of means across multiple groups simultaneously, avoiding inflation of Type I error from multiple t-tests. Reliability theory quantifies the probability that a system functions over time, essential in engineering and quality management.

6.1 One-way ANOVA (CRD)

ONE-WAY ANOVA TABLE

SS BETWEEN (SSA)

$\text{SSA} = \sum_i n_i(\bar{x}_i - \bar{x})^2$

SS WITHIN (SSE)

$\text{SSE} = \sum_{i,j}(x_{ij} - \bar{x}_i)^2$

F-RATIO

$F = \frac{\text{MSA}}{\text{MSE}} = \frac{\text{SSA}/(k-1)}{\text{SSE}/(N-k)}$

DEGREES OF FREEDOM

Between: k−1; Within: N−k; Total: N−1

DECISION RULE

Reject H_0 ($\mu_1=\mu_2=\ldots=\mu_k$) if $F > F_{crit}$

TWO-WAY ANOVA (RBD)

Add SS for blocks; df=(r−1)(c−1) for error

1 One-way ANOVA

Three groups (k=3, N=12): SSA=60, SSE=90. Test at α=0.05 (F_{crit}=4.26).

SOLUTION

1. $df_1=k-1=2$; $df_2=N-k=9$. MSA=60/2=30; MSE=90/9=10
2. F=MSA/MSE=30/10=3
3. F=3 < F_{crit}=4.26 → Fail to reject H_0. No significant difference in means.

F=3 < 4.26. Fail to reject H_0. Means are not significantly different.

6.2 Reliability Theory

RELIABILITY FORMULAS

RELIABILITY FUNCTION

$R(t) = P(T > t) = 1 - F(t)$

HAZARD RATE (FAILURE RATE)

$h(t) = \frac{f(t)}{R(t)}$

EXPONENTIAL MODEL

$R(t) = e^{-\lambda t}$; $h(t) = \lambda$ (constant)

MTTF (EXP. MODEL)

$\mathrm{MTTF} = \int_0^\infty R(t)\,dt = \frac{1}{\lambda}$

WEIBULL MODEL

$R(t) = e^{-(t/\beta)^\alpha}$; MTTF=$\beta\Gamma(1 + 1/\alpha)$

SERIES SYSTEM

$R_s = \prod_i R_i(t)$

1 Exponential reliability

A component has failure rate λ=0.01/hour. Find R(100h), MTTF, and h(t).

SOLUTION

1. $R(100) = e^{-0.01\times100} = e^{-1} \approx 0.3679 \approx 36.8\%$
2. $\mathrm{MTTF} = 1/\lambda = 1/0.01 = 100$ hours
3. Hazard rate h(t)=λ=0.01/hour (constant — exponential distribution)

R(100)≈36.8%; MTTF=100h; h(t)=0.01/h (constant)

Module 6 — Key Formulas

ANOVA F-ratio

MSA/MSE

Reliability R(t)

$e^{-\lambda t}$ (exponential)

MTTF

$1/\lambda$ (exponential)

MODULE 7

Multiple Regression & Non-parametric Tests

Multiple regression extends simple regression to multiple predictors, essential for real-world data analysis. Non-parametric tests make no assumption about the population distribution and are used when data is ordinal, skewed, or sample sizes are small.

7.1 Multiple Linear Regression

MULTIPLE REGRESSION

MODEL EQUATION

$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$

MATRIX FORM

$\hat{\mathbf{y}} = X\beta$; $\hat{\beta} = (X^TX)^{-1}X^T\mathbf{y}$

R² (EXPLAINED VAR.)

$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} \in [0, 1]$

ADJUSTED R²

$R^2_{adj} = 1 - \frac{\text{SSE}/(n-k-1)}{\text{SST}/(n-1)}$

OVERALL F-TEST

H_0: all $\beta_i=0$. $F = \frac{SSR/k}{SSE/(n-k-1)}$

INDIVIDUAL T-TEST

$t = \hat{b}_j/\text{SE}(\hat{b}_j),\ df = n-k-1$

1 Interpret multiple regression output

Fitted model: $\hat{y} = 5 + 3x_1 + 2x_2$, $R^2=0.85$. Interpret β_1.

SOLUTION

1. $\beta_1=3$: holding x_2 constant, a unit increase in x_1 increases $\hat{y}$ by 3 units
2. $R^2=0.85$: 85% of variance in y is explained by x_1 and x_2 — a strong model

$\beta_1=3$ (partial effect); $R^2=85\%$ variance explained

7.2 Non-parametric Tests

NON-PARAMETRIC TEST SUMMARY

MANN-WHITNEY U TEST

Two independent samples; ranks combined; H_0: equal distributions. $U = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$

WILCOXON SIGNED-RANK

Paired samples; rank absolute differences; H_0: symmetric about 0

KRUSKAL-WALLIS

Non-parametric one-way ANOVA. Rank all data; $H = \frac{12}{N(N+1)} \sum \frac{R_i^2}{n_i} - 3(N+1) \sim \chi^2_{k-1}$

SIGN TEST

Count + and − signs; binomial distribution with p=0.5

WHEN TO USE NON-PARAMETRIC

Small samples; non-normal data; ordinal data; outliers present

ADVANTAGE

Distribution-free; robust to outliers; works with ranks/ordinal data

1 Mann-Whitney U test

Group A: 3,5,7,9 (n_1=4). Group B: 2,4,6,8 (n_2=4). Ranks combined (ascending): 2(B),3(A),4(B),5(A),6(B),7(A),8(B),9(A). R_1 = sum of ranks for A.

SOLUTION

1. Ranks of Group A: 3→rank2, 5→rank4, 7→rank6, 9→rank8. R_1=2+4+6+8=20
2. $U = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 16 + 10 - 20 = 6$
3. Critical value for n_1=n_2=4, α=0.05: U_crit=1. U=6 > 1 → Fail to reject H_0

U = 6; Fail to reject H_0 (no significant difference between groups)

2 Kruskal-Wallis test

3 groups, N=12. Sum of ranks: R_1=22, R_2=30, R_3=26 (each n=4). Compute H.

SOLUTION

1. $H = \frac{12}{12(13)} \left[\frac{22^2}{4} + \frac{30^2}{4} + \frac{26^2}{4}\right] - 3(13)$
2. $= \frac{1}{13}\left[\frac{484+900+676}{4}\right] - 39 = \frac{1}{13} \cdot \frac{2060}{4} - 39 = \frac{515}{13} - 39 = 39.62 - 39 = 0.62$
3. df=k−1=2. χ²_crit(0.05,2)=5.991. H=0.62 < 5.991 → Fail to reject H_0

H=0.62 < 5.991. No significant difference among groups.

Module 7 — Key Formulas

Regression R^2

$1 - \mathrm{SSE}/\mathrm{SST}$

Mann-Whitney U

$n_1 n_2 + n_1(n_1+1)/2 - R_1$

Kruskal-Wallis H

$\frac{12}{N(N+1)} \sum R_i^2/n_i - 3(N+1)$

BMAT202L — Master Formula Sheet

Modules 1–3: Descriptive Stats, Probability & Distributions

VARIANCE

$\sigma^2 = \Sigma x^2/n - \bar{x}^2$

PEARSON R

$n\Sigma xy - \Sigma x \Sigma y$ over $\sqrt{D_x D_y}$

BAYES'

$P(A|B) = P(B|A)P(A)/P(B)$

BINOMIAL

E=np; Var=npq

POISSON

E=Var=λ; $P = e^{-\lambda}\lambda^k/k!$

NORMAL Z-SCORE

$Z = (X - \mu)/\sigma$

Modules 4–5: Inference & Testing

CLT

$\bar{X} \sim N(\mu, \sigma^2/n)$

95% CI

$\bar{x} \pm 1.96\sigma/\sqrt{n}$

Z-TEST

$Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$

T-TEST

$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Χ² TEST

$\Sigma(O - E)^2/E$

F-TEST

s_1^2/s_2^2

Modules 6–7: ANOVA, Reliability & Non-parametric

ANOVA F

MSA/MSE

R(T) EXPONENTIAL

$e^{-\lambda t}$

MTTF

$1/\lambda$

REGRESSION R²

$1 - \mathrm{SSE}/\mathrm{SST}$

MANN-WHITNEY U

$n_1 n_2 + n_1(n_1 + 1)/2 - R_1$

KRUSKAL-WALLIS H

χ^2_{k-1} approx

BMAT202L — Complete

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